

Corpus callosal reference ranges: systematic review of methodology of biometric chart construction and measurements obtained

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KEYWORDS: corpus callosum; fetal biometry; fetal growth; fetal MRI

CONTRIBUTION

What are the novel findings of this work?

Our systematic review demonstrates substantial heterogeneity in methods and final biometric values of the length of the fetal corpus callosum in the existing literature.

What are the clinical implications of this work?

The heterogeneity in study methods may explain discrepancies between the reported charts. Clearly defined and consistent methodology is therefore necessary to critically evaluate such population differences. The use of uniform methodology of the highest quality is essential in order to define a ‘short’ corpus callosum and provide appropriate parental counseling.

ABSTRACT

Objective Adequate reference ranges of size of the corpus callosum (CC) are necessary to improve characterization of CC abnormalities and parental counseling. The objective of this study was to evaluate the methodology used in studies developing references charts for CC biometry.

Methods We conducted a systematic review of studies on fetal CC biometry using a set of predefined quality criteria of study design, statistical analysis and reporting methods. We included observational studies whose primary aim was to create ultrasound or magnetic resonance imaging charts for CC size in a normal population of fetuses. Studies were scored against a predefined set of independently agreed

methodological criteria, and an overall quality score was given for each study.

Results Twelve studies met the inclusion criteria. Quality scores ranged between 17.4% and 95.7%. The greatest potential for bias was noted for the following items: sample selection and sample-size calculation, as only 17% of the studies were population-based and had consecutive or random recruitment of patients and with a justification of the sample size; number of measurements obtained for CC biometry, as only 17% of the studies performed more than one measurement per fetus and per scan; and description of characteristics of the study population, as only 8% of the studies clearly reported a minimum dataset of demographic characteristics.

Conclusions Our review demonstrates substantial heterogeneity in methods and final biometric values of the fetal CC across the evaluated studies. The use of uniform methodology of the highest quality is essential in order to define a ‘short’ CC and provide appropriate parental counseling. © 2023 The Authors. *Ultrasound in Obstetrics & Gynecology* published by John Wiley & Sons Ltd on behalf of International Society of Ultrasound in Obstetrics and Gynecology.

INTRODUCTION

The corpus callosum (CC) is an interhemispheric structure that enables the integration of motor, sensory and cognitive functions between the two cerebral hemispheres, which is essential for cognitive activities¹. The fetal

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CC is considered a sensitive marker of normal brain development and maturation², and fetuses with partial or complete agenesis of the CC, even if isolated, are at increased risk of abnormal neurodevelopment^{3,4}.

Many morphological pathologies of the CC have been described, and it is recognized that, in some pathologies, the CC is morphologically 'normal' but may be too short, too thin or too thick^{5–8}. These CC pathologies are generally described as one of the following: complete agenesis, characterized by total absence of the CC; hypogenesis of the CC, also known as partial agenesis and characterized by partial absence of the CC; hypoplasia of the CC, characterized by a fully formed CC but thinner than expected for age and sex of the individual; or dysgenesis of the CC, which includes a thickened and/or abnormally shaped CC⁹. Thus, there is great heterogeneity in the nomenclature and definition of CC anomalies other than complete agenesis¹⁰. Availability of adequate terminology and reference ranges of the size of the CC is mandatory to better characterize partial agenesis of the CC and improve parental counseling.

Developing reference ranges is a complex process, as has been demonstrated by the development of standard biometric charts for other fetal parameters and the ongoing debates regarding the selection of the most appropriate growth chart^{11–13}. It is likely that differences in study design and data analysis contribute to discrepancies among the different charts for CC biometry. Suboptimal methodology used to produce a fetal CC size chart may affect the ability to discriminate between healthy and affected fetuses. The aim of this systematic review was to evaluate the methodology used to develop different reference charts for CC biometry.

METHODS

This systematic review of observational studies was conducted and reported following the checklist proposed by the MOOSE group¹⁴. A search in *PubMed* was conducted for the period 1968–2022 using combinations of relevant medical subject heading terms and keywords, including 'fetal', 'corpus callosum' and 'length', to identify studies of fetal CC biometry (Figure S1). The search was not restricted according to study design or methodology, but only articles published in English or French were considered. The titles of all identified studies were reviewed for relevance based on whether development of fetal CC biometry charts was the goal of the study. The online search was complemented by review of the references of all retrieved full articles.

Eligible studies were prospective or retrospective case series or cohort studies of a normal fetal population undergoing assessment using ultrasound or magnetic resonance imaging (MRI) with the goal of developing standard reference charts for the fetal CC. Articles were excluded if the primary aim did not involve the construction of CC size charts, if postmortem measurements were used to build size charts or if they were a systematic review or meta-analysis.

One reviewer (R.C.), a specialist in fetal medicine, screened the title and abstract of all identified citations and selected potentially eligible studies. The same reviewer independently assessed the full-text version of eligible studies.

The quality of the included studies was assessed using a list of methodological quality criteria based on available published research (Table 1)¹⁵. These quality criteria were divided into three domains: study design, statistical methods and reporting methods.

The studies were assessed by two reviewers (R.C., L.J.S.) against each criterion within the checklist and were classified as having high or low risk of bias. Disagreements were resolved either by consensus or through consultation with a third reviewer (D.G.). The overall quality score was calculated for each study as the percentage of criteria classified as low risk of bias out of the total number of quality criteria.

To assess imaging quality assurance, the following items were evaluated: measurement blinding, standardization of the measurement technique(s), storage of images, independent image review, image scoring and monitoring of intra- and/or interobserver variation.

Statistical analysis was performed using the Statistical Package for Social Sciences (SPSS) version 25.0 (IBM Corporation, Armonk, NY, USA). To represent the evolution of the length of the CC during gestation, we plotted either the values obtained with the regression polynomial equations provided by the authors or the means of the length of the CC at each gestational age (GA). This review was not registered and a protocol was not prepared. Our research received no specific grant or funding.

RESULTS

Characteristics and overall quality score of studies

Twelve publications from seven countries published between 1993 and 2018 and that met the study criteria were identified^{1,16–26}. Measurement of the fetal CC was performed using two-dimensional (2D) ultrasound (seven studies), three-dimensional (3D) ultrasound (three studies) or MRI (two studies). Eight (66.7%) studies had a prospective design and four (33.3%) had a retrospective design. The median sample size of the participating women was 274 (range, 67–2950), while the median number of scans was 327 (range, 68–2950). Eight (66.7%) studies reported only on biometric parameters of the CC, whereas the remaining four (33.3%) included a combination of brain biometrics. Ten (83.3%) of the studies reported only one CC measurement per examination and, in six (50.0%) studies, measurements were performed by several practitioners. Pregnancy dating was based on crown–rump length (CRL) in four (33.3%) studies, last menstrual period (LMP) in two (16.7%) studies, LMP confirmed by CRL in five (41.7%) studies and LMP confirmed by early ultrasound in one (8.3%) study.

Table 1 Methodological quality criteria¹⁵

Domain	Low risk of bias	High risk of bias
1. Study design		
1.1 Design	Clearly described as cross-sectional or longitudinal	Not reported or mixture of cross-sectional and longitudinal data
1.2 Sample selection	Population-based study in which there are attempts to identify and clearly define populations from a specific geographical area; from this underlying population, women are selected consecutively or randomly	Not population-based, arbitrary recruitment or not reported
1.3 Number of scans per fetus	Each fetus was included only once	Some fetuses were included more than once
1.4 Inclusion/exclusion criteria	It is clearly stated that women at high risk of pregnancy complications and those with abnormal outcome were excluded, i.e. best efforts were made to include cases with 'normal' outcome. As a minimum, study population excluded: multiple pregnancy, fetuses with congenital structural or chromosomal anomaly, cases of fetal death, women with disorder that may affect fetal growth (at least should specify exclusion of women with pre-existing hypertension, diabetes mellitus, renal disease and smoking), pregnancy complications (at least pre-eclampsia) and pregnancies conceived by ART	Study population included both low-risk and high-risk pregnancies, women with abnormal outcome or fetuses/women with characteristics in left column. Exclusions which would have a direct effect on the estimated percentiles, such as fetuses found to be LGA or SGA at birth
1.5 Sample-size calculation	<i>A-priori</i> determination/calculation of sample size and justification	Lack of <i>a-priori</i> sample-size determination/calculation and justification
1.6 Data collection	Prospective study and ultrasound data collected specifically for constructing charts of fetal size or growth	Retrospective study, data not collected specifically for constructing charts of fetal size or growth (e.g. use of routinely collected data) or unclear type of data
1.7 Pregnancy dating	Clearly described; known LMP and regular menstrual cycles prior to pregnancy and sonogram before 11 weeks demonstrating CRL that corroborates LMP dates (within how many days unspecified) or sonogram between 11 and 14 weeks demonstrating CRL	Not described clearly or GA assessment performed after 14 weeks or did not include sonographic verification
1.8 GA at inclusion calculation	GA calculated precisely to the day	Truncation of GA to number of 'completed weeks'
2. Statistical methods		
2.1 Number of measurements taken for each biometric variable	More than one measurement per fetus per scan	Single measurement or not specified
2.2 Statistical methods	Clearly described and identified	Not clearly described and identified
2.3 Assessment of increasing variability of data with gestation	Performed	Not performed
2.4 Assessment of goodness-of-fit of model	Goodness-of-fit testing of models reported	Goodness-of-fit testing of models not reported
2.5 Scatterplot of data with fitted percentiles superimposed	Study included scatterplots of data with percentiles superimposed	Study did not include scatterplots of data with percentiles superimposed
2.6 Change in reference percentiles across gestation	Smooth change	Not smooth change
2.7 Methods used to estimate GA-specific reference intervals for CC measurements	Mean and SD model, smoothed crude percentiles or LMS method ¹⁶	Inadequate
3. Reporting methods		
3.1 Characteristics of study population	Presented in a table or clearly described and includes minimum dataset of age, weight, height or body mass index and parity	Not presented in a table or not clearly described or did not contain minimum dataset
3.2 Description of number approached/enrolled	Described	Not described

Continued over.

Table 1 Continued

Domain	Low risk of bias	High risk of bias
3.3 Ultrasound or MRI equipment used	Clearly specified	Not clearly specified
3.4 Number of participating practitioners	Reported	Not reported
3.5 Description of measurement technique(s)	Sufficient and unambiguous details of technique used for CC measurement provided, including imaging plane and caliper placement method	Sufficient and unambiguous details of technique used for CC measurement not provided
3.6 Quality control measures	Should include assessment of intraobserver and interobserver variability, image review, image storing and image scoring	Does not contain quality control measures
3.7 Reporting of mean and SD of each measurement and sample size for each week of gestation	Presented in a table or clearly described	Not presented in a table or not clearly described
3.8 Reporting of regression equations for mean (and SD if relevant) for each measurement	Reported	Not reported

ART, assisted reproductive technology; CC, corpus callosum; CRL, crown–rump length; GA, gestational age; LGA, large-for-gestational age; LMP, last menstrual period; MRI, magnetic resonance imaging; SGA, small-for-gestational age.

Table 2 Characteristics of studies and overall quality scores

Study	Country	Women included	Scans performed	GA (weeks)	Design	Scan type	Measurements	Quality score (%)
Malinger (1993) ²³	Israel	101	101	18–42	Prosp	2D-US	CC length	21.74
Achiron (2001) ¹⁸	Israel	255	255	16–36	Prosp	2D-US	CC length and thickness	47.83
Zhang (2009) ¹	China	622	549	16–39	Retro	2D-US	CC length	17.39
Tilea (2009) ²⁶	France	589	589	26–40	Retro	MRI	CC length and other neurological parameters	52.17
Harreld (2011) ²⁰	USA	67	68	18.5–37.7	Retro	MRI	CC length	52.17
Rizzo (2011) ²⁵	Italy	361	361	18–32	Prosp	3D-US	CC length and other neurological parameters	52.17
Goldstein (2011) ¹⁹	Israel	252	252	16–33	Prosp	2D-US	CC length	60.87
Araujo Júnior (2012) ²¹	USA	293	293	20–33.6	Prosp	3D-US	CC length and area	65.22
Pashaj (2013) ²⁴	Albania	466	466	17–41.6	Prosp	3D-US	CC length and other neurological parameters	60.87
Cignini (2014) ¹⁶	Italy	2950	2950	19–37.6	Prosp	2D-US	CC length	95.65
Koning (2017) ²²	Netherlands	157	685	22–32	Prosp	2D-US	CC length and other neurological parameters	56.52
Tsur (2019) ¹⁷	Israel	255	255	21–37.6	Retro	2D-US	CC length	30.43

Data are given as *n* or range, unless stated otherwise. Only first author is given for each study. 2D, two-dimensional; 3D, three-dimensional; CC, corpus callosum; GA, gestational age at measurement; MRI, magnetic resonance imaging; Prosp, prospective; Retro, retrospective; US, ultrasound.

The overall quality score for each study is presented in Table 2 and Figure 1. The breakdown of scores per study and per quality criterion is presented in Table S1. The proportion of studies with a low risk of bias for each quality criterion in the study design, statistical methods and reporting methods domains are presented in Figure 2a–c and Table S2.

Study design

The highest risk of bias was noted for sample selection and sample-size calculation, as only 16.7% of studies were population-based and had consecutive or random recruitment of patients and a justification of the sample size.

GA was calculated precisely to the day in only 25.0% of studies. Regarding the inclusion and/or exclusion criteria, a high risk of bias was noted in 66.7% of the studies, which included both low-risk and high-risk pregnancies

but did not describe inclusion/exclusion criteria and/or included fetuses with other brain abnormalities. Details of all exclusion criteria used are presented in Table 3.

Statistical methods

The highest risk of bias was noted in terms of the number of measurements obtained for CC biometry, as more than one measurement per fetus and per scan was performed in only 16.7% of studies.

Reporting methods

The highest risk of bias was noted in terms of the description of characteristics of the study population, as only 8.3% of studies included a table or a clear description of a minimum dataset of demographic characteristics of the included women. Only in 25.0% of studies were mean and SD of CC measurements presented in a table or

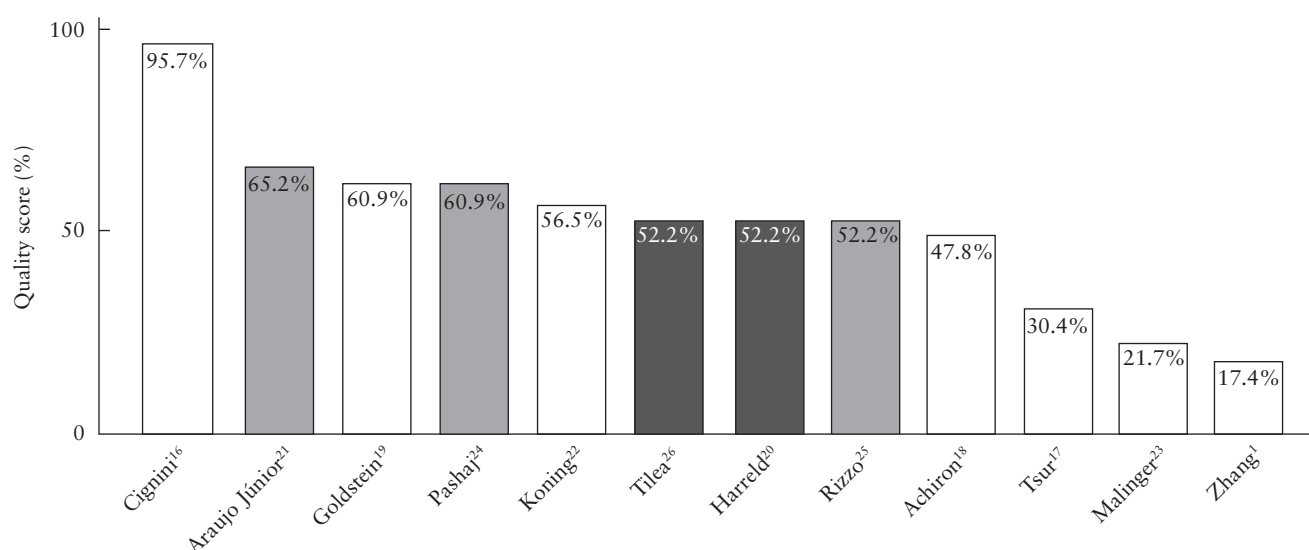


Figure 1 Overall quality score of studies included in systematic review. Only first author is given for each study. □, two-dimensional ultrasound; ■, three-dimensional ultrasound; ■, magnetic resonance imaging.

clearly described, including the sample size for each week of gestation. Only seven (58.3%) studies presented the 5th or 10th percentile for CC length^{16,19,21–24,26}.

Ultrasound quality assurance measures

Details of the quality assurance measures are presented in Table 4. Measurement of CC parameters was standardized in 83.3% of the studies, and inter- and intraobserver variation was evaluated in 66.7% and 58.3% of the studies, respectively. Both inter- and intraobserver agreement was excellent, with intraclass correlation coefficients ranging from 0.75 to 0.99 and from 0.93 to 0.99, respectively^{16–18,20–22,25,26}. Only 25.0% of sonographers were blinded to the fetal condition.

Measurement of CC

In most studies, CC length was measured as the maximum length measurable in the midsagittal plane using ultrasound or MRI, from the most anterior aspect of the genu to the most posterior aspect of the splenium using a straight rostrocaudal line. No study detailed how the calipers were placed.

Comparison between charts

Table 5, Figure 3 and Figures S2–S4 present a comparison between charts for fetal CC length. Significant discrepancies were observed across gestation between charts in the reported studies. When ultrasound and MRI studies were separated, significant discrepancies were also observed, regardless of the imaging technique.

Partial agenesis of CC

Partial agenesis of the CC occurs in at least 0.1% of the general population and is of clinical interest because CC

anomalies are frequently associated with other structural malformations and genetic diseases as well as a risk of neurodevelopmental delay in 25–30% of cases^{27–29}. Table 6 and Figure S5 present a comparison between the different lower thresholds of CC length measurements reported in the literature. In five (41.7%) studies, a lower threshold to define partial CC agenesis was not defined or was not reported. During the second trimester, at 24 weeks, the 5th percentile threshold to define partial CC agenesis ranged between 23 mm and 26 mm. During the third trimester, at 32 weeks, the 5th percentile ranged between 34 mm and 36 mm.

DISCUSSION

Our systematic review demonstrates substantial heterogeneity in biometric values of the fetal CC across studies as well as methods used, which is reflected by quality scores ranging between 17% and 96%. This is in accordance with other reviews of general fetal biometry^{14,30,31}. Differences in study quality may explain discrepancies between the reported charts. Clearly defined and consistent methodology is therefore necessary to critically appraise such population differences.

A large proportion of studies (66.7%) did not provide a clear description of inclusion/exclusion criteria or included both low-risk and high-risk pregnancies. The aim of fetal CC biometry charts is to depict how the CC grows under optimal conditions. As many maternal and fetal conditions, such as intrauterine growth restriction, cerebral ischemia, infection and brain congenital abnormality, may impact on fetal CC growth²³, precise definitions of inclusion and exclusion criteria are mandatory for the external validation of fetal CC biometric charts. This is particularly relevant in MRI studies, in which inclusion bias is likely, as MRI is not usually performed in routine practice.

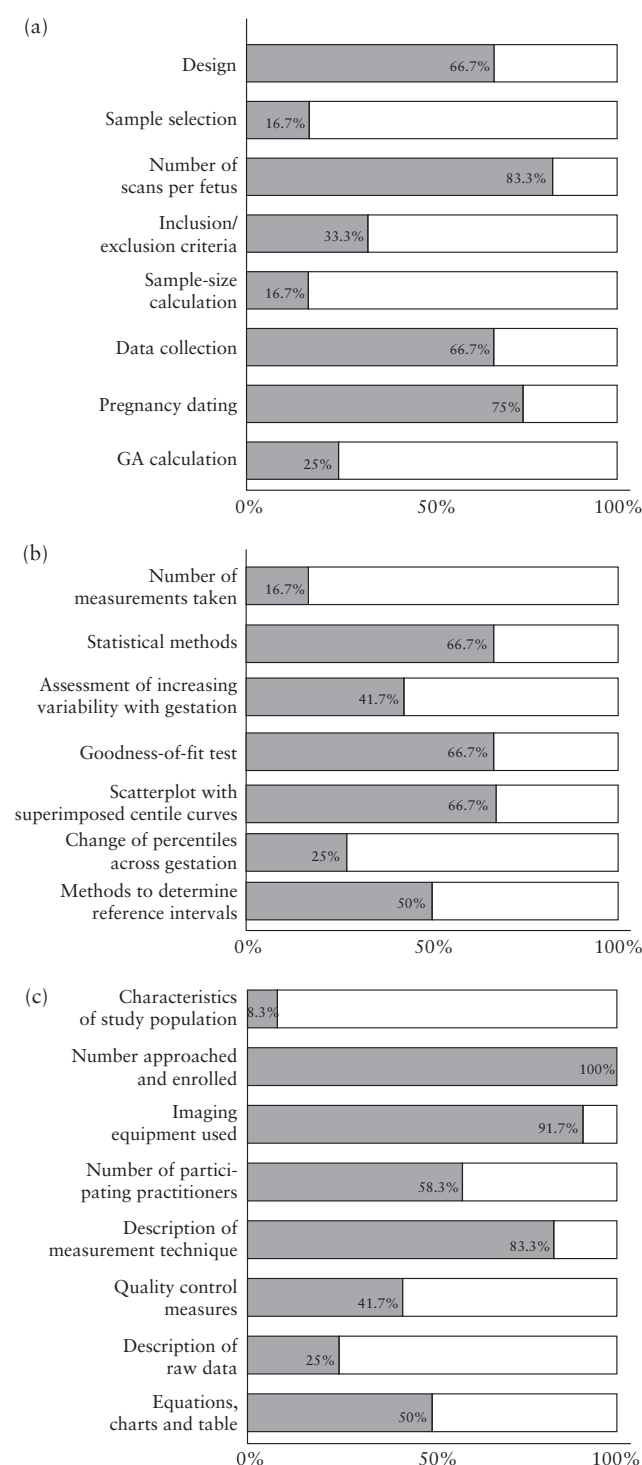


Figure 2 Methodological quality of included study in study-design (a), statistical-methods (b) and reporting-methods (c) domains. GA, gestational age.

Several different pregnancy dating strategies were encountered in this systematic review. There is now robust evidence that early sonographic determination of GA is more reliable than LMP alone³². It has also been argued that sonographic biometry alone may be marginally superior to LMP confirmed by ultrasound, although the differences are small³³.

One-third of the studies included in this systematic review reported only the raw measurement of the fetal

CC, which represents a high risk of bias. A consensus has been reached in recent years regarding the appropriate methods for statistical modeling and data presentation^{16,19,34}. The presentation of raw measurement data only is not clinically useful or informative. Both the median and variance should be modeled as a function of GA, and goodness-of-fit testing should demonstrate that these curves describe accurately the structure of raw data³⁴.

This systematic review focused on several imaging modalities for fetal CC measurement, thus the use of different techniques may explain some of the discrepancies among fetal charts reported here. Little is known about which type of CC measurement is more reliable. Miguelote *et al.*³⁵ demonstrated a reliable method of measuring CC length using 3D imaging reconstruction. However, a comparison of 2D, 3D and four-dimensional volume-contrast imaging suggests that 2D imaging has superior quality, while 3D imaging, although easy to perform, does not overcome all the limitations of 2D imaging³⁶.

We have demonstrated heterogeneity of existing fetal CC biometric charts, particularly before 24 weeks' and after 32 weeks' gestation. Many studies did not present the number of cases evaluated for each GA. It is possible that the difference in CC measurement before 24 weeks and after 32 weeks is explained by a low number of cases included in these GA windows. We hypothesize that few data were available outside the typical GA window for second-trimester anatomy scans for ultrasound-based studies or around 31 weeks for MRI studies. Establishing the growth pattern of the fetal CC is important, since it has been shown that the CC may be an indicator of normal brain development and maturation³⁷. Differences in CC size in neonates and adults have been associated with abnormalities in cognitive abilities^{38–40}. Moreover, CC agenesis and dysgenesis are frequently accompanied by other central nervous system abnormalities^{41,42}. Although direct visualization and measurement of the CC have not yet become an integral part of the routine anomaly scan⁴³, indirect signs of CC anomalies should prompt a more detailed assessment of the structure. Early identification of CC dysgenesis can be valuable for pregnancy management, delivery and neonatal care. While a short CC may indicate the presence of a neurodevelopmental disorder, it can also represent a normal developmental variation. Inclusion of other landmarks as reference points or further detailed work-up by advanced MRI tools, such as MRI tractography, may be helpful in that regard^{44,45}.

To our knowledge, no study has yet demonstrated, with high clinical evidence, higher accuracy of MRI compared with ultrasound in the measurement of CC length. Harreld *et al.*²⁰ compared their MRI measurements of the CC with those reported in ultrasound studies. They demonstrated that measurements of CC length by ultrasound and MRI were statistically equivalent.

Our systematic review has several strengths. First, the quality criteria used were based on the best available evidence. Second, the use of a quality score in a percentage

Table 3 Exclusion criteria considered in reported studies

Parameter	All studies (n = 12)	2D-US studies (n = 7)	MRI studies (n = 2)	3D-US studies (n = 3)
Hypertensive disorders	4 (33.3)	1 (14.3)	0 (0)	3 (100)
Diabetes mellitus	4 (33.3)	1 (14.3)	0 (0)	3 (100)
Multiple pregnancy	5 (41.7)	1 (14.3)	1 (50.0)	3 (100)
Genetic disorders	4 (33.3)	1 (14.3)	0 (0)	3 (100)
Congenital malformation	5 (41.7)	1 (14.3)	1 (50.0)	3 (100)
No dating confirmation	8 (66.7)	4 (57.1)	1 (50.0)	3 (100)
History of congenital malformation	1 (8.3)	1 (14.3)	0 (0)	0 (0)
Intrauterine growth restriction	3 (25.0)	1 (14.3)	1 (50.0)	1 (33.3)
Podalic presentation	1 (8.3)	1 (14.3)	0 (0)	0 (0)

Studies using listed exclusion criteria were considered low risk with regard to these criteria. 2D, two-dimensional; 3D, three-dimensional; MRI, magnetic resonance imaging; US, ultrasound.

Table 4 Imaging quality assurance measures

Parameter	All studies (n = 12)	2D-US studies (n = 7)	MRI studies (n = 2)	3D-US studies (n = 3)
Measurement blinding	3 (25.0)	1 (14.3)	2 (100)	0 (0)
Standardization	10 (83.3)	5 (71.4)	2 (100)	3 (100)
Images stored	5 (41.7)	1 (14.3)	2 (100)	2 (66.7)
Image scoring	3 (25.0)	1 (14.3)	0 (0)	2 (66.7)
Intraobserver variation	7 (58.3)	4 (57.1)	1 (50.0)	2 (66.7)
Interobserver variation	8 (66.7)	4 (57.1)	2 (100)	2 (66.7)

Studies employing listed quality assurance measures were considered low risk with regard to these criteria. 2D, two-dimensional; 3D, three-dimensional; MRI, magnetic resonance imaging; US, ultrasound.

Table 5 Measurement values of fetal corpus callosal length across gestation in studies included in systematic review

Measurement	Malinger ²³	Achiron ¹⁸	Zhang ¹	Tilea ²⁶	Harreld ²⁰	Rizzo ²⁵	Goldstein ¹⁹	Araujo Junior ²¹	Pashaj ²⁴	Cignini ¹⁶	Koning ²²	Tsur ¹⁷
Scans (n)	101	255	549	589	68	361	252	293	466	2950	685	255
Imaging technique	2D-US	2D-US	2D-US	MRI	MRI	3D-US	2D-US	3D-US	3D-US	2D-US	2D-US	2D-US
Parameter	Mean	Mean	Mean	Median	Mean	Mean	Median	Median	Mean	Mean	Mean	Mean
GA												
16 weeks	—	3.75	9.4	—	—	—	—	—	—	—	—	—
17 weeks	—	6.24	12	—	—	—	11.45	—	—	—	—	—
18 weeks	—	12.5	14.0	—	16.5	23.1	—	—	9.3	—	—	—
19 weeks	16.9	15.8	16.6	—	18.8	24.2	16.75	—	13.9	18.8	—	—
20 weeks	—	19.0	18.7	—	21.0	25.3	—	18.5	17.5	21.0	—	—
21 weeks	20.6	20.4	22.2	—	23.0	26.4	21.7	20.8	20.6	23.2	—	23.30
22 weeks	—	22.4	23.9	—	25.0	27.6	25.8	23.0	23.3	25.3	26.4	25.62
23 weeks	23.3	24.5	25.6	—	26.9	28.7	27.65	25.1	25.8	27.3	—	26.02
24 weeks	—	27.6	26.0	—	28.7	29.8	29.3	27.1	28.0	29.2	—	30.03
25 weeks	29.8	29.7	29.0	—	30.3	30.9	31.2	28.9	29.9	31.1	—	31.57
26 weeks	—	31.4	31.8	31.7	31.9	32.0	33.3	30.7	31.7	32.8	34.3	31.20
27 weeks	33.7	34.3	33.8	33.4	33.4	33.1	34.5	32.3	33.3	34.5	—	32.28
28 weeks	—	34.4	34.6	34.9	34.8	34.2	—	33.8	34.7	36.0	—	35.51
29 weeks	25.8	36.4	35.5	36.2	36.1	35.3	37.9	35.2	35.9	37.4	—	36.95
30 weeks	—	38.3	36.5	37.3	37.4	36.4	—	36.5	37.0	38.7	—	38.01
31 weeks	36.8	37.3	37.7	38.3	38.5	37.6	39.8	37.7	37.9	39.9	—	39.21
32 weeks	—	40.4	38.2	39.1	39.5	38.7	—	38.7	38.7	40.9	41.6	37.50
33 weeks	39.1	38.5	38.5	39.8	40.4	—	—	39.6	39.3	41.8	—	41.57
34 weeks	—	42.5	39.8	40.4	41.3	—	—	—	39.9	42.6	—	41.88
35 weeks	40.6	45.6	42.6	40.9	42.0	—	—	—	40.3	43.2	—	43.17
36 weeks	—	44.0	44.7	41.3	42.6	—	—	—	40.6	43.7	—	41.34
37 weeks	41.9	44.7	45.0	41.7	43.2	—	—	—	40.9	44.0	—	43.10
38 weeks	—	—	44.2	42.1	—	—	—	—	41.1	—	—	—
39 weeks	43	—	43.2	42.3	—	—	—	—	41.2	—	—	—
40 weeks	—	—	—	42.6	—	—	—	—	—	—	—	—

Only first author is given for each study. 2D, two-dimensional; 3D, three-dimensional; GA, gestational age; MRI, magnetic resonance imaging; US, ultrasound.

format allowed an objective rather than empirical assessment of quality. However, we acknowledge some limitations. This review was restricted to studies written in English or French, and studies of normative fetal CC biometry available in other languages may have been missed. Another limitation is that only one reviewer

(R.C.) performed the data abstraction not blinded to the origin and authors of the included studies. Moreover, this review did not evaluate the morphology of the CC, and it is known that differences in the shape of the CC, as well as its size, have been associated with abnormalities in cognitive ability in neonates and adults^{38–40}.

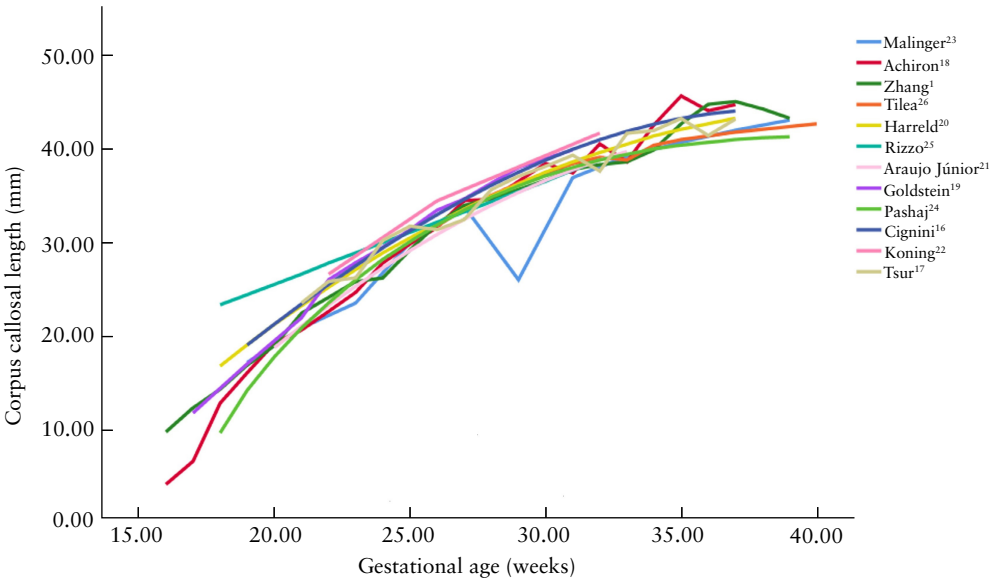


Figure 3 Length of fetal corpus callosum across gestation in all studies included in systematic review. Only first author is given for each study.

Table 6 Lower threshold of corpus callosal length used to define partial agenesis of corpus callosum in studies included in systematic review

Measurement	Malinger ²³	Tilea ²⁶	Goldstein ¹⁹	Araujo Júnior ²¹		Pashaj ²⁴	Cignini ¹⁶		Koning ²²
Scans (n)	101	589	252	293		466	2950		685
Imaging technique	2D-US	MRI	2D-US	3D-US		3D-US	2D-US		2D-US
Parameter	Mean	Median	Median	Median		Mean	Mean		Mean
Lower threshold	–1 SD	10 th p	10 th p	5 th p	10 th p	5 th p	5 th p	10 th p	–1 SD
GA									
17 weeks	—	—	8.08	—	—	—	—	—	—
18 weeks	—	—	—	—	—	6	—	—	—
19 weeks	14.5	—	12.81	—	—	10.5	16.89	17.38	—
20 weeks	—	—	—	14.23	15.17	14	18.67	19.19	—
21 weeks	16.2	—	20.26	16.56	17.50	19.6	20.67	21.23	—
22 weeks	—	—	22.54	18.76	19.70	22.0	22.59	23.19	25.13
23 weeks	20.3	—	24.95	20.85	21.79	24.2	24.42	25.06	—
24 weeks	—	—	26.28	22.83	23.77	26.1	26.18	26.85	—
25 weeks	27.5	—	28.34	24.69	25.63	27.8	27.83	28.55	—
26 weeks	—	25.4	31.16	26.43	27.37	29.3	29.39	30.15	32.47
27 weeks	31.3	28.1	30.50	28.05	28.99	30.7	30.85	31.65	—
28 weeks	—	30.4	—	29.56	30.50	31.9	32.20	33.03	—
29 weeks	23	32.2	34.50	30.96	31.90	33.4	33.44	34.31	—
30 weeks	—	33.6	—	32.23	33.17	33.8	34.55	35.46	—
31 weeks	35.4	34.8	34.44	33.39	34.33	34.6	35.55	36.50	—
32 weeks	—	35.7	—	34.44	35.38	35.2	36.41	37.40	39.37
33 weeks	34.8	36.4	—	35.36	36.30	35.8	37.14	38.17	—
34 weeks	—	37.0	—	—	—	36.2	37.73	38.80	—
35 weeks	34.2	37.4	—	—	—	36.5	38.18	39.29	—
36 weeks	—	37.7	—	—	—	36.7	38.48	39.62	—
37 weeks	38.4	37.9	—	—	—	36.9	38.62	39.80	—
38 weeks	—	38.0	—	—	—	37.0	—	—	—
39 weeks	38.8	38.1	—	—	—	37.0	—	—	—
40 weeks	—	38.1	—	—	—	—	—	—	—

Only first author is given for each study. 2D, two-dimensional; 3D, three-dimensional; MRI, magnetic resonance imaging; p, percentile; US, ultrasound.

Future research should develop a fetal CC biometry chart taking into consideration all the quality criteria mentioned in this study, employing a prospective, multicenter study design, including patients randomly, regardless of fetal or maternal condition, and using reproducible and clearly detailed methodology of fetal CC measurement by several practitioners. Before such a study is available, we recommend using the methodology described by Cignini *et al.*¹⁶ to measure the fetal CC based on the quality scores results. Using a transabdominal ultrasound probe, the CC has to be visualized in a midsagittal plane as an anechoic structure delimited by two echogenic lines: superiorly by the sulcus of the CC and the cingulate gyrus and inferiorly by the cavum septi pellucidi, cavum vergae and lateral ventricles. The length can be measured from the most anterior aspect of the genu to the most posterior aspect of the splenium using a straight rostrocaudal line.

Conclusion

This systematic review has demonstrated heterogeneity of design across studies of fetal CC biometry charts. The use of uniform methodology of the highest quality is essential in order to define a 'short' CC and provide appropriate parental counseling.

REFERENCES

- Zhang H-c, Yang J, Chen Z-p, Ma X-y. Sonographic study of the development of fetal corpus callosum in a Chinese population. *J Clin Ultrasound* 2009; 37: 75–77.
- Goodyear PWA, Bannister CM, Russell S, Rimmer S. Outcome in prenatally diagnosed fetal agenesis of the corpus callosum. *Fetal Diagn Ther* 2001; 16: 139–145.
- Moutard ML, Kieffer V, Feingold J, Kieffer F, Lewin F, Adamsbaum C, Gélort A, Campistol i Plana J, van Bogaert P, André M, Ponsot G. Agenesis of corpus callosum: prenatal diagnosis and prognosis. *Childs Nerv Syst* 2003; 19: 471–476.
- Shevell MI. Clinical and diagnostic profile of agenesis of the corpus callosum. *J Child Neurol* 2002; 17: 896–900.
- Wegiel J, Kuchna I, Nowicki K, Imaki H, Wegiel J, Marchi E, Ma SY, Chauhan A, Chauhan V, Bobrowicz TW, De Leon M, Louis LAS, Cohen IL, London E, Brown WT, Wisniewski T. The neuropathology of autism: defects of neurogenesis and neuronal migration, and dysplastic changes. *Acta Neuropathol* 2010; 119: 755–770.
- Romaniello R, Arrigoni F, De Salvo P, Bonaglia MC, Panzeri E, Bassi MT, Parazzini C, Righini A, Borgatti R. Long-term follow-up in a cohort of children with isolated corpus callosum agenesis at fetal MRI. *Ann Clin Transl Neurol* 2021; 8: 2280–2288.
- Hinkley LBN, Marco EJ, Findlay AM, Honma S, Jeremy RJ, Strominger Z, Bukshpun P, Wakahiro M, Brown WS, Paul LK, Barkovich AJ, Mukherjee P, Nagarajan SS, Sherr EH. The role of corpus callosum development in functional connectivity and cognitive processing. *PLoS One* 2012; 7: e39804.
- Paul LK. Developmental malformation of the corpus callosum: a review of typical callosal development and examples of developmental disorders with callosal involvement. *J Neurodev Disord* 2011; 3: 3–27.
- Paul LK, Brown WS, Adolphs R, Tyska JM, Richards LJ, Mukherjee P, Sherr EH. Agenesis of the corpus callosum: genetic, developmental and functional aspects of connectivity. *Nat Rev Neurosci* 2007; 8: 287–299.
- Mahallati H, Sotiriadis A, Celestin C, Millischer AE, Sonigo P, Grevent D, O'Gorman N, Bahi-Buisson N, Attié-Bitach T, Ville Y, Salomon LJ. Heterogeneity in defining fetal corpus callosal pathology: systematic review. *Ultrasound Obstet Gynecol* 2021; 58: 11–18.
- Liauw J, Mayer C, Albert A, Fernandez A, Hutcheon JA. Which chart and which cut-point: deciding on the INTERGROWTH, World Health Organization, or Hadlock fetal growth chart. *BMC Pregnancy Childbirth* 2022; 22: 25.
- Papageorgiou AT, Ohuma EO, Altman DG, Todros T, Ismail LC, Lambert A, Jaffer YA, Bertino E, Gravett MG, Purwar M, Noble JA, Pang R, Victora CG, Barros FC, Carvalho M, Salomon LJ, Bhutta ZA, Kennedy SH, Villar J. International standards for fetal growth based on serial ultrasound measurements: the Fetal Growth Longitudinal Study of the INTERGROWTH-21st Project. *Lancet* 2014; 384: 869–879.
- Kiserud T, Piaggio G, Carroli G, Widmer M, Carvalho J, Neerup Jensen L, Giordano D, Cecatti JG, Abdel Aleem H, Talegawkar SA, Benachi A, Diemert A, Tshetu Kitoto A, Thinkhamrop J, Lumbiganon P, Tabor A, Kriplani A, Gonzalez Perez R, Hecher K, Hanson MA, Gülmezoglu AM, Platt LD. The World Health Organization Fetal Growth Charts: A Multinational Longitudinal Study of Ultrasound Biometric Measurements and Estimated Fetal Weight. *PLoS Med* 2017; 14: 1002220.
- Stroup DF, Berlin JA, Morton SC, Olkin I, Williamson GD, Rennie D, Moher D, Becker BJ, Sipe TA, Thacker SB. Meta-analysis of observational studies in epidemiology: a proposal for reporting. Meta-analysis Of Observational Studies in Epidemiology (MOOSE) group. *JAMA* 2000; 283: 2008–2012.
- Ioannou C, Talbot K, Ohuma E, Sarris I, Villar J, Conde-Agudelo A, Papageorgiou AT. Systematic review of methodology used in ultrasound studies aimed at creating charts of fetal size. *BJOG* 2012; 119: 1425–1439.
- Cignini P, Padula F, Giorlandino M, Brutti P, Alfò M, Giannarelli D, Mastrandrea ML, D'Emidio L, Vacca L, Aloisi A, Giorlandino C. Reference charts for fetal corpus callosum length: a prospective cross-sectional study of 2950 fetuses. *J Ultrasound Med* 2014; 33: 1065–1078.
- Tsur A, Weisz B, Rosenblat O, Shai D, Derazne E, Stevenson DK, Achiron R, Katorza E. Personalized charts for the fetal corpus callosum length. *J Matern Fetal Neonatal Med* 2019; 32: 3931–3938.
- Achiron R, Achiron A. Development of the human fetal corpus callosum: a high-resolution, cross-sectional sonographic study. *Ultrasound Obstet Gynecol* 2001; 18: 343–347.
- Goldstein I, Tamir A, Reece AE, Weiner Z. Corpus callosum growth in normal and growth-restricted fetuses. *Prenat Diagn* 2011; 31: 1115–1119.
- Harreld JH, Bhore R, Chason DP, Twickler DM. Corpus callosum length by gestational age as evaluated by fetal MR imaging. *AJNR Am J Neuroradiol* 2011; 32: 490–494.
- Araujo Júnior E, Visentainer M, Simioni C, Ruano R, Nardoza LM, Moron AF. Reference values for the length and area of the fetal corpus callosum on 3-dimensional sonography using the transfrontal view. *J Ultrasound Med* 2012; 31: 205–212.
- Koning IV, Roelants JA, Groenenberg IAL, Vermeulen MJ, Willemsen SP, Reiss IKM, Govaert PP, Steegers-Theunissen RPM, Dudink J. New ultrasound measurements to bridge the gap between prenatal and neonatal brain growth assessment. *AJNR Am J Neuroradiol* 2017; 38: 1807–1813.
- Malinge G, Zakut H. The corpus callosum: normal fetal development as shown by transvaginal sonography. *AJR Am J Roentgenol* 1993; 161: 1041–1043.
- Pashaj S, Merz E, Wellek S. Biometry of the fetal corpus callosum by three-dimensional ultrasound. *Ultrasound Obstet Gynecol* 2013; 42: 691–698.
- Rizzo G, Pietrolucci ME, Capponi A, Arduini D. Assessment of corpus callosum biometric measurements at 18 to 32 weeks' gestation by 3-dimensional sonography. *J Ultrasound Med* 2011; 30: 47–53.
- Tilea B, Alberti C, Adamsbaum C, Armoogum P, Oury JF, Cabrol D, Sebag G, Kalifa G, Garbel C. Cerebral biometry in fetal magnetic resonance imaging: new reference data. *Ultrasound Obstet Gynecol* 2009; 33: 173–181.
- Santo S, D'antonio F, Homfray T, Rich P, Piliu G, Bhide A, Thilaganathan B, Papageorgiou AT. Counseling in fetal medicine: agenesis of the corpus callosum. *Ultrasound Obstet Gynecol* 2012; 40: 513–521.
- Shwe WH, Schlatterer SD, Williams J, du Plessis AJ, Mulkey SB. Outcome of Agenesis of the Corpus Callosum Diagnosed by Fetal MRI. *Pediatr Neurol* 2022; 135: 44–51.
- Shevell MI. Clinical and Diagnostic Profile of Agenesis of the Corpus Callosum. *J Child Neurol* 2002; 17: 896–900.
- Di Mascio D, Khalil A, Rizzo G, Kasprian G, Caulo M, Manganaro L, Odibo AO, Flacco ME, Giancotti A, Buca D, Liberati M, Timor-Tritsch IE, D'Antonio F. Reference ranges for fetal brain structures using magnetic resonance imaging: systematic review. *Ultrasound Obstet Gynecol* 2022; 59: 296–303.
- Donadono V, Cavallaro A, Roberts NW, Ioannou C, Papageorgiou AT, Napolitano R. A Systematic Review of Methodology Used in Studies Aimed at Creating Charts of Fetal Brain Structures. *Diagnostics (Basel)* 2021; 11: 916.
- Gardosi J. Dating of pregnancy: time to forget the last menstrual period. *Ultrasound Obstet Gynecol* 1997; 9: 367–368.
- Mongelli M, Wilcox M, Gardosi J. Estimating the date of confinement: ultrasonographic biometry versus certain menstrual dates. *Am J Obstet Gynecol* 1996; 174: 278–281.
- Altman DG, Chitty LS. Design and analysis of studies to derive charts of fetal size. *Ultrasound Obstet Gynecol* 1993; 3: 378–384.
- Miguelote RF, Vides B, Santos RF, Matias A, Sousa N. Feasibility and reproducibility of transvaginal, transabdominal, and 3D volume reconstruction sonography for measurement of the corpus callosum at different gestational ages. *Fetal Diagn Ther* 2012; 31: 19–25.
- Piliu G, Ghi T, Carletti A, Segata M, Perolo A, Rizzo N. Three-dimensional ultrasound examination of the fetal central nervous system. *Ultrasound Obstet Gynecol* 2007; 30: 233–245.
- Sztrihla L. Spectrum of corpus callosum agenesis. *Pediatr Neurol* 2005; 32: 94–101.
- Volpe P, Paladini D, Resta M, Stanziano A, Salvatore M, Quarantelli M, De Robertis V, Buonadonna AL, Caruso G, Gentile M. Characteristics, associations and outcome of partial agenesis of the corpus callosum in the fetus. *Ultrasound Obstet Gynecol* 2006; 27: 509–516.
- Liu F, Cao S, Liu J, Du Z, Guo Z, Ren C. Ultrasound measurement of the corpus callosum and neural development of premature infants. *Neural Regen Res* 2013; 8: 2432–2440.
- Nosarti C, Rushe TM, Woodruff PWR, Stewart AL, Rifkin L, Murray RM. Corpus callosum size and very preterm birth: relationship to neuropsychological outcome. *Brain* 2004; 127: 2080–2089.

41. Hetts SW, Sherr EH, Chao S, Gobuty S, Barkovich AJ. Anomalies of the corpus callosum: an MR analysis of the phenotypic spectrum of associated malformations. *AJR Am J Roentgenol* 2006; 187: 1343–1348.
42. Barkovich AJ, Norman D. Anomalies of the corpus callosum: correlation with further anomalies of the brain. *AJR Am J Roentgenol* 1988; 151: 171–179.
43. Malinger G, Paladini D, Haratz KK, Monteagudo A, Pilu GL, Timor-Tritsch IE. ISUOG Practice Guidelines (updated): sonographic examination of the fetal central nervous system. Part 1: performance of screening examination and indications for targeted neurosonography. *Ultrasound Obstet Gynecol* 2020; 56: 476–484.
44. Millischer A-E, Grevent D, Sonigo P, Bahi-Buisson N, Desguerre I, Mahallati H, Bault J-P, Quibel T, Couderc S, Moutard M-L, Julien E, Dangouloff V, Bessieres B, Malan V, Attie T, Salomon L-J, Boddaert N. Feasibility and Added Value of Fetal DTI Tractography in the Evaluation of an Isolated Short Corpus Callosum: Preliminary Results. *AJNR Am J Neuroradiol* 2022; 43: 132–138.
45. Kasprian G, Brugger PC, Schöpf V, Mitter C, Weber M, Hainfellner JA, Prayer D. Assessing prenatal white matter connectivity in commissural agenesis. *Brain* 2013; 136: 168–179.
46. Cole TJ. Using the LMS method to measure skewness in the NCHS and Dutch National height standards. *Ann Hum Biol* 1989; 16: 407–419.

SUPPORTING INFORMATION ON THE INTERNET

The following supporting information may be found in the online version of this article:



Figure S1 MOOSE flowchart summarizing study search.

Figure S2 Length of fetal corpus callosum across gestation in two-dimensional ultrasound studies included in systematic review.

Figure S3 Length of fetal corpus callosum across gestation in three-dimensional ultrasound studies included in systematic review.

Figure S4 Length of fetal corpus callosum across gestation in magnetic resonance imaging studies included in systematic review.

Figure S5 Lower threshold of length of fetal corpus callosum across gestation in studies included in systematic review.

Table S1 Risks of bias of studies included in systematic review

Table S2 Subanalysis of methodological quality of included studies according to imaging technique used



Intervalos de referencia del cuerpo calloso: revisión sistemática de la metodología de elaboración de gráficas biométricas y de las mediciones obtenidas

RESUMEN

Objetivo. Los intervalos de referencia adecuados del tamaño del cuerpo calloso (CC) son necesarios para mejorar la caracterización de las anomalías del CC y el asesoramiento a los padres. El objetivo de este estudio fue evaluar la metodología empleada en los estudios de elaboración de gráficas de referencia para la biometría del CC.

Métodos. Se realizó una revisión sistemática de los estudios sobre biometría del CC del feto utilizando un conjunto de criterios de calidad predefinidos de diseño del estudio, análisis estadístico y métodos de presentación de informes. Se incluyeron estudios observacionales con el objetivo principal de crear gráficas de ecografía o resonancia magnética para el tamaño del CC en una población normal de fetos. Los estudios se puntuaron respecto a un conjunto predefinido de criterios metodológicos acordados de forma independiente y a cada estudio se le asignó una puntuación en cuanto a su calidad.

Resultados. Doce estudios cumplieron los criterios de inclusión. Las puntuaciones de calidad se situaron entre el 17,4% y el 95,7%. El mayor potencial de sesgo se observó en los siguientes aspectos: selección de la muestra y cálculo del tamaño de la muestra, ya que sólo el 17% de los estudios estaban basados en una población y contaban con un reclutamiento consecutivo o aleatorio de pacientes y con una justificación del tamaño de la muestra; número de mediciones obtenidas para la biometría del CC, ya que sólo el 17% de los estudios realizaron más de una medición por feto y por exploración; y descripción de las características de la población de estudio, ya que sólo el 8% de los estudios reportaron claramente sobre un conjunto mínimo de datos de características demográficas.

Conclusiones. Esta revisión demuestra una heterogeneidad sustancial en los métodos y los valores biométricos finales del CC fetal entre los estudios evaluados. El uso de una metodología uniforme de la máxima calidad es esencial para definir un CC ‘corto’ y proporcionar un asesoramiento adecuado a los padres.

胼胝体参考范围：对生物测量图表构建方法和测量结果的系统评价

目的 为了更好地确定胼胝体（CC）异常的特征，改进对家长的咨询指导，有必要对胼胝体的大小设定适当的参考范围。本研究旨在评估为 CC 生物测量制定参考图表的研究中使用的方法。

方法 我们采用一套预先确定的研究设计、统计分析和报告方法质量标准，对有关胎儿 CC 生物测量的研究进行了系统性回顾。我们纳入的观察性研究的主要目的是为正常胎儿群体的 CC 大小制作超声或磁共振成像图表。我们根据预先确定的一套独立商定的方法学标准对研究进行评分，并对每项研究给出总体质量评分。

结果 共 12 项研究符合纳入标准。质量得分为 17.4% 至 95.7% 不等。其中以下几项最有可能出现偏差：样本选择和样本量计算，因为只有 17% 的研究以人群为基础，曾连续或随机招募患者，并对 CC 生物测量的样本量和测量次数作出解释；只有 17% 的研究对每个胎儿和每次扫描进行了一次以上的测量，并对研究人群特征进行了描述；并且只有 8% 的研究明确报告了最低限度的人口统计学特征数据集。

结论 我们的回顾显示，在所评估的研究中，胎儿 CC 的生物测量方法和最终测量值存在很大的异质性。要界定何谓胼胝体“短小”，并为家长提供适用的咨询指导，必须使用最高质量的统一方法。